

Cybersecurity Project Progress Status Tracking

Project Overview

This document tracks the progress and deliverables for the cybersecurity project, including system design, security assessment, mitigation implementation, threat modeling, and final presentation.

Progress Milestones

20% Completion – Project Selection & Architecture

Objectives

- Select the project/application.
- Set up the development environment.
- Ensure the application is operational.
- Complete system design and architecture documentation.

Deliverables

- Project scope and objectives.
 - System architecture diagram.
 - Component inventory.
 - Data flow documentation.
 - Environment setup guide.
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40% Completion – Security Analysis

Objectives

- Identify security vulnerabilities.
- Define trust boundaries.
- Analyze interfaces and attack surfaces.
- Document secure data handling requirements.

Deliverables

- Vulnerability assessment report.
- Trust boundary analysis.
- Security requirements documentation.
- Risk identification and mitigation recommendations.

Key Areas

- Authentication risks
 - Authorization weaknesses
 - Input validation issues
 - Data exposure risks
 - Network security concerns
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60% Completion – Mitigation Implementation

Objectives

Implement or document security controls and mitigation strategies.

Deliverables

- Authentication implementation.
- Authorization controls.
- Audit logging and monitoring.
- HTTPS/TLS configuration.
- Encryption for sensitive data.
- Secrets management strategy.
- Security testing results.

Example Security Controls

1. Authentication
 - Multi-factor authentication (MFA)
 - Secure password storage
 2. Authorization
 - Role-Based Access Control (RBAC)
 - Principle of Least Privilege
 3. Audit & Monitoring
 - Security event logging
 - User activity tracking
 4. Encryption
 - Encryption in transit (HTTPS/TLS)
 - Encryption at rest for sensitive data
 5. Secrets Management
 - Vault-based secret storage
 - Key rotation procedures
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80% Completion – Threat Modeling

Objectives

Complete comprehensive threat modeling documentation.

Deliverables

- Threat model report.
- Data Flow Diagram (DFD).
- Asset inventory.
- Threat actor analysis.
- Attack trees.
- Risk assessment.

Threat Modeling Components

Actors

- External attackers
- Internal users
- Administrators
- Third-party services

Assets

- User data
- Credentials
- Application services
- Databases
- APIs

Threat Analysis

- STRIDE Analysis
 - Spoofing
 - Tampering
 - Repudiation
 - Information Disclosure
 - Denial of Service
 - Elevation of Privilege
- DREAD Risk Assessment
 - Damage Potential
 - Reproducibility
 - Exploitability
 - Affected Users
 - Discoverability

Attack Trees

- Unauthorized access
 - Data exfiltration
 - Privilege escalation
 - Service disruption
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100% Completion – Final Presentation

Objectives

Prepare and deliver the final project presentation.

Presentation Topics

1. Project Overview
 - Objectives
 - Scope
 - Architecture
2. Security Assessment
 - Vulnerabilities identified
 - Risk analysis
3. Security Improvements
 - Implemented controls
 - Mitigation strategies
4. Threat Modeling
 - DFDs
 - STRIDE analysis
 - DREAD scoring
 - Attack trees
5. Key Learnings
 - Challenges encountered
 - Lessons learned
 - Security best practices
6. Demonstration
 - Before and after security posture
 - Implemented security features

Audience “Aha” Moments

- Real-world impact of identified vulnerabilities.
- Demonstration of how mitigations reduce risk.
- Visualization of attack paths using attack trees.
- Measurable security improvements were achieved during the project.

Progress Tracking Table

Milestone	Status	Completion
Project Selection & Architecture	Complete	20%
Security Analysis	In Progress	40%
Mitigation Implementation	Pending	60%
Threat Modeling	Pending	80%
Final Presentation	Pending	100%

Conclusion

This project follows a structured cybersecurity lifecycle beginning with system design, progressing through security assessment and mitigation implementation, and concluding with comprehensive threat modeling and presentation of findings. Successful completion demonstrates both technical security improvements and a strong understanding of secure system design principles.